STA553-Data Visualization

Instructor: Cheng Peng

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1 Contact Information

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- Office Hours:

Day	Time	Location
Tuesday/Thursday	3:00 PM-5:00 PM	R107, UNA25/ Zoom
By appointment		

- **ZOOM Link**: Available on the course web page and D2L as well.
- **Course Web Page**: https://pengdsci.github.io/STA553VIZ/
- **Classroom**: 25 University Avenue 120

2 Modality

This course will be a distance education (DE) class, taught asynchronously. A well-designed, comprehensive course website will host all class materials. Assignments and projects will be submitted through the D2L Dropbox. There will be no in-person attendance requirements.

3 Course Description

This course examines the foundational design principles of data and information visualization, emphasizing what to visualize, why particular design strategies are effective, and how visual representations can be crafted to support compelling storytelling and meaningful pattern discovery. Core topics include the purposeful use of fundamental visualization elements (marks, visual channels, facets, and guides) along with exploratory data visualization, introductory visual analytics, real-time process visualization, dashboard design, and the ethical dimensions of visual communication. The course also introduces contemporary technologies and AI-driven tools that facilitate the development of interactive visualizations capable of revealing insights within complex datasets.

4 Learning Objectives and Outcomes

4.1 Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Design and create statistical graphics grounded in the fundamental principles of data visualization.

2. Conduct exploratory data analysis using a variety of tools and techniques.
3. Critically evaluate existing visualizations and propose meaningful improvements and refinements.
4. Design and evaluate interactive visualization tools for usability and effectiveness.
5. Develop web-based interactive visualizations.

5 Logistics and Required Materials

- **Textbooks (optional):** No required textbook for this class.
- **Class Notes:** Class notes will be provided.
- **Computational Tools:**
  - Primary Programming languages and Software: R and Tableau Public
  - Platforms: RStudio, Github, Shiny
  - Typesetting: LaTex and Markdown
- **Coverage:** See the list of tentative topics

6 Policies for Using AI

You are encouraged to use AI as a tool to support and enrich your learning. Throughout the course, I will demonstrate effective strategies for collaborating with AI when developing and refining your visual design ideas. However, AI should not be used as a substitute for your own work; it cannot independently satisfy the requirements of course assignments. The policies below outline the appropriate and acceptable uses of AI in this class.

- **Allowed Uses**: You may use AI tools to help debug code, brainstorm ideas, explain concepts, or refine your visual design.
- **Prohibited Uses**: Do not use AI to generate complete visualizations. Your visualization must meet the design requirements and adhere to the design principles covered in the class.
- **You Are Responsible**: All submitted work must be your own. Your code should be clearly written, well commented, and organized in a manner that allows others to easily understand its logic and structure. While AI-generated code may reflect the design philosophies of its developers, you are responsible for verifying that any such code aligns with your own understanding and produces the intended results prior to submission on D2L.
- **Be Transparent**: Include a brief note with each assignment stating which AI tools you used and how you used them.
- **Ask When Unsure**: If you are uncertain whether a specific use of AI is appropriate, please ask the instructor before submitting.

7 Assessments

7.1 Assessment Components

The course grade consists of the following components in this 5-week summer session. All assignments and the term project will use real-world or near-real-world datasets drawn from my teaching data repository (https://pengdsci.github.io/datasets/) or other online sources, including the UCI Machine Learning Repository(https://archive.ics.uci.edu/) and Kagle (https://www.kaggle.com/).

- Weekly assignments (68%) (17% percent each assignment)

  - *Assignment 1*: Focuses on base R graphics and `ggplot2`, integrating core visual design principles with an emphasis on graphical theme design within the `ggplot2` framework.
  - *Assignment 2*: Focuses on interactive visualization using `plotly` and related graphical libraries, with an emphasis on designing graphical layouts, popups, and tooltips.
  - *Assignment 3*: Involves creating interactive maps to visualize spatial data, with an emphasis on incorporating popup subplots and tables to display insights from other dimensions.
  - *Assignment 4*: Focuses on Tableau dashboard design using business process monitoring data.
- Class attendance and participation (10%)
- Term project - report and presentation (22%)
  - Focuses on R Shiny app design strategies and their application in a Shiny dashboard.

7.2 Grading Rubrics

- **Clarity of Insight Communication**: The visualization successfully conveys its central insight with precision and minimal cognitive effort. Patterns, trends, or comparisons are immediately apparent without requiring excessive explanation, ensuring the primary narrative is accessible to the intended audience.
- **Appropriateness of Visual Encoding**: Marks, channels (e.g., position, length, color), and chart types are chosen to accurately and honestly represent the underlying data. Encodings align with the nature of the data (categorical, sequential, diverging) and support rather than distort the intended analytical comparison.
- **Effective Use of Interaction and Annotation**: Interactive elements (e.g., tooltips, filtering) enhance data exploration without obscuring core insights. Annotations, labels, and captions are used judiciously to guide interpretation, highlight key findings, and provide necessary context.
- **Aesthetic Quality**: Visual appeal including consistent typography, thoughtful color palettes, and uncluttered layout—contributes to usability and accessibility. Aesthetics are considered only insofar as they support clarity and do not introduce distortion or unnecessary complexity.

7.3 Grade Scales

The final course grade will be calculated based on the above components. A letter grade will be assigned according to the following scale.

Grade	Quality Points	Percentage Equivalents	Interpretation
A	4.00	[93%, 100%]	Superior graduate attainment
A-	3.67	[90%, 93%)	
B+	3.33	[86%, 90%)	Satisfactory graduate attainment
B	3.00	[83%, 86%)	
B-	2.67	[80%, 83%)	
C+	2.33	[76%, 80%)	Attainment below graduate expectations
C	2.00	[73%, 76%)	
C-	1.67	[70%, 73%)	
F	0	$< 70\%$	Failure

D grades are not used. Refer to the Graduate Catalog for the description of NG (No Grade), W, & other grades.

8 Class Policies

8.1 Attendance and Participation

Class attendance is mandatory. Active participation in class discussions is required and will factor into the final course grade.

8.2 Late Homework and Assignments

Late assignments will be accepted; however, they will be subject to a modest penalty deduction.

9 Tentative Topics

- Logistics and Technical Tools
  - **Equipment:** Laptop with reliable internet connection
  - **Primary Language:** R, along with several graphical libraries
  - **Notebook:** R Markdown
  - **Markup Languages:** HTML, CSS, JavaScript (no prior experience assumed; will be introduced)
  - **Platforms & Web Servers (freely available online):** GitHub, RPubs, Tableau Public, SAS Studio OnDemand
  - Rendering interactive visualizations
  - Setting up a personal webpage and creating an online resume via GitHub
- Ethics & Design Principles of DataViz:
  - **Core Ethical Principle:** Convey honest and relevant information from data
  - **Ethical Dimensions of Data Visualization**
  - **Basic Design Principles:** Balance clarity, information density, structure, and visual appeal
  - **Best Practices:** Guide attention, reduce clutter, show truth, design with purpose, and more
- R & Base R Graphics
  - **Basic R Coding Knowledge:** Syntax, control flow, user-defined functions
  - **R Graphics Devices:** Layout design and output configuration
  - **Enhancing Plots:** Adding graphical features to existing plots using secondary R graphics functions
  - **General Guideline:** Avoid 3D graphics whenever possible
- Data Processing for DataViz
  - **Base R Functions** for data processing
  - **Tidyverse Library**
  - **Overview of Relational Databases** and basic SQL syntax
  - **Table Structures:** Long and wide tables for different graphical comparisons
- ggplot Revisited
  - **ggplot Review**
  - **Graphical Theme Designs**
  - **Interactive ggplot:** Using the wrapper function `ggplotly()`
- Interactive Plot with plotly library
  - **Syntax:** `plot_ly()`
  - **Layout Design:** Analogous to themes in ggplot
  - **Comparisons:** `plot_ly()` vs. `ggplotly()`
  - **Popups & Tooltips:** Enhancing the effectiveness of visuals
- Interactive Thematic Maps: Visualizing Spatial Information
  - **Types of Thematic Maps**
  - **Interactive Map Libraries in R**
  - **Adding Spatial Information** to existing interactive maps
- Advanced Maps: Annotation Graphic Functions for Maps
  - **Map Structure:** Shapefiles, coordinate reference systems, and attribute data
  - **Tooltips:** Viewing additional feature information

  - **Popups:** Static and interactive subplots
- Getting Started with Tableau Interactive Graphics
  - **Loading Data Sources**
  - **View Type Selection**
  - **Mark Cards:** Refining analytic graphics
  - **Ensembling Graphics:** Building Tableau visuals through modular components
- Dashboard Design and Tableau Dashboard
  - **Principles of Effective Dashboard Design**
  - **Operational Dashboards**
  - **Analytical Dashboards**
  - **Tableau Storyboards**
- Getting Started With ShinyApps: UI and Server Design
  - **R Shiny Graphics Structure**
  - **Front-End UI Design:** Structuring user interfaces with layouts, widgets, and dynamic UI elements
  - **Back-End Compute Design:** Building reactive logic, managing state, and optimizing performance
- Confirmatory and Exploratory Data Analysis with Shiny Apps
  - **Analytic Shiny Graphics** for repetitive analytical tasks
  - **Concepts of Dynamic Visual Analytics**
- R Flexdashboard: Dashboard Design Strategies
  - **R Markdown Extension** for static and dynamic dashboards
  - **Designing Dashboard Building Blocks**
  - **Visual Design Principles:** "Less is more" and balanced design
- RShinyDashboard and Multiple FlexDashboard
  - **R Shiny Dashboard:** An R Shiny extension for highly interactive web applications
  - **Multiple Flexdashboards:** An analogy to Tableau storyboards

10 WCU Distance Education Statements

- **Minimum Student Preparation and Prerequisites**: Students are expected to possess basic computer literacy skills, including the ability to browse the internet, send and receive emails with attachments, and create and manage files. Students must be able to install software on the computer they use for this course; therefore, the use of a personal computer is required. Work-issued devices often have administrative restrictions ("locks") that prevent the installation of necessary software (such as proctoring tools or specific browser extensions). Consequently, work-restricted computers are not suitable for this course.

- **Minimum Technical Skills**: To succeed in this course, students must be proficient in navigating the D2L (Brightspace) learning management system, including uploading assignments to Dropbox, posting to Discussion Boards, and accessing feedback. Students must also be proficient in using West Chester University email, Zoom for video conferencing, and the Microsoft Office Suite (Word, PowerPoint, and Excel) for creating and editing course documents.

- **Required Hardware and Software**:

  - **Hardware**: Access to a modern desktop or laptop computer (Windows 10/11 or macOS 11+) with a working webcam and microphone is required. Chromebooks, iPads, and mobile devices are not fully compatible with all WCU systems and should not be relied upon as a primary device.

  - **Internet**: A reliable, high-speed broadband internet connection is required.

  - **Software**: Microsoft Office 365: West Chester University provides Microsoft Office 365 (Word, Excel, PowerPoint, etc.) to all active students at no cost. You can install these applications on up to five personal devices by logging into office.com(https://www.office.com/) with your WCU

credentials and clicking "Install Office."

  - **Web Browser**: Google Chrome or Mozilla Firefox are the preferred browsers for D2L.
  - **PDF Reader**: Adobe Acrobat Reader (free version) or similar.

- **Technical Support Contacts**: WCU Information Services & Technology (IS&T) Help Desk:
  - Primary Support for: Email, Passwords, Office 365, WiFi, and Login issues.
  - Phone: 610-436-3350
  - Email: helpdesk@wcupa.edu
  - Web: ServiceNow Support Portal(https://wcupaprod.service-now.com/)
  - Location: Anderson Hall, Room 119

11 University Policies and Resources

11.1 ACADEMIC & PERSONAL INTEGRITY

It is the responsibility of each student to adhere to the university's standards for academic integrity. Violations of academic integrity include any act that violates the rights of another student in academic work, that involves misrepresentation of your own work, or that disrupts the instruction of the course. Other violations include (but are not limited to): cheating on assignments or examinations; plagiarizing, which means copying any part of another's work and/or using ideas of another and presenting them as one's own without giving proper credit to the source; selling, purchasing, or exchanging of term papers; falsifying of information; and using your own work from one class to fulfill the assignment for another class without significant modification. Proof of academic misconduct can result in the automatic failure and removal from this course. For questions regarding Academic Integrity, the No-Grade Policy, Sexual Harassment, or the Student Code of Conduct, students are encouraged to refer to the Department Graduate Handbook, the Graduate Catalog, the Ram's Eye View, and the University website at www.wcupa.edu.

11.2 ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

West Chester University is committed to providing equitable access to the full WCU experience for Golden Rams of all abilities. Students should contact the Office of Educational Accessibility (OEA) to establish accommodations if they have had accommodations in the past or if they believe they may be eligible for accommodations due to a disability, whether or not it may be readily apparent. There is no deadline for disclosing to OEA or for requesting to use approved accommodations in a given course. However, accommodations can only be applied to future assignments or exams; that is, they can't be applied retroactively. Please share your letter from OEA as soon as possible so that we can discuss accommodations.

If you have concerns related to disability discrimination, please contact the university's ADA Coordinator in the Office of Diversity, Equity, and Inclusion or 610-436-2433.

11.3 EXCUSED ABSENCES POLICY

Students are advised to carefully read and comply with the excused absences policy, including absences for university-sanctioned events, contained in the WCU Graduate Catalog. In particular, please note that the "responsibility for meeting academic requirements rests with the student," that this policy does not excuse students from completing required academic work, and that professors can require a "fair alternative" to attendance on those days that students must be absent from class in order to participate in a University-Sanctioned Event.

11.4 REPORTING INCIDENTS OF SEXUAL VIOLENCE

West Chester University and its faculty are committed to assuring a safe and productive educational environment for all students. In order to comply with the requirements of Title IX of the Education

Amendments of 1972 and the University's commitment to offering supportive measures in accordance with the new regulations issued under Title IX, the University requires faculty members to report incidents of sexual violence shared by students to the University's Title IX Coordinator. The only exceptions to the faculty member's reporting obligation are when incidents of sexual violence are communicated by a student during a classroom discussion, in a writing assignment for a class, or as part of a University-approved research project. Faculty members are obligated to report sexual violence or any other abuse of a student who was, or is, a child (a person under 18 years of age) when the abuse allegedly occurred to the person designated in the University Protection of Minors Policy. Information regarding the reporting of sexual violence and the resources that are available to victims of sexual violence is set forth at: Diversity, Equity, and Inclusion at WCU website

11.5 INCLUSIVE LEARNING ENVIRONMENT AND ANTI-RACIST STATEMENT

Diversity, equity, and inclusion are central to West Chester University's mission as reflected in our Mission Statement, Values Statement, Vision Statement and Strategic Plan: Pathways to Student Success. We disavow racism and all actions that silence, threaten, or degrade historically marginalized groups in the U.S. We acknowledge that all members of this learning community may experience harm stemming from forms of oppression including but not limited to classism, ableism, heterosexism, sexism, Islamophobia, anti-Semitism, and xenophobia, and recognize that these forms of oppression are compounded by racism.

Our core commitment as an institution of higher education shapes our expectation for behavior within this learning community, which represents diverse individual beliefs, backgrounds, and experiences. Courteous and respectful behavior, interactions, and responses are expected from all members of the University. We must work together to make this a safe and productive learning environment for everyone. Part of this work is recognizing how race and other aspects of who we are shape our beliefs and our experiences as individuals. It is not enough to condemn acts of racism. For real, sustainable change, we must stand together as a diverse coalition against racism and oppression of any form, anywhere, at any time.

Resources for education and action are available through WCU's Office for Diversity, Equity, and Inclusion (ODEI), DEI committees within departments or colleges, the student ombudsperson, and centers on campus committed to doing this work (e.g., Dowdy Multicultural Center, Center for Women and Gender Equity, and the Center for Trans and Queer Advocacy).

Guidance on how to report incidents of discrimination and harassment is available at the University's Office of Diversity, Equity and Inclusion.

11.6 EMERGENCY PREPAREDNESS

All students are encouraged to sign up for the University's free WCU ALERT service, which delivers official WCU emergency text messages directly to your cell phone. For more information, visit WCU Alert. To report an emergency, call the Department of Public Safety at 610-436-3311.

11.7 ELECTRONIC MAIL POLICY

It is expected that faculty, staff, and students activate and maintain regular access to University provided e-mail accounts. Official university communications, including those from your instructor, will be sent through your university e-mail account. You are responsible for accessing that mail to be sure to obtain official University communications. Failure to access will not exempt individuals from the responsibilities associated with this course.